

TØ 5 – Mikroøkonomi 2 – 2024

Problem 1

Consider the following normal-form game G , where $x > 8$:

$1 \backslash 2$	P_2	Q_2	R_2	S_2
P_1	4,4	$x,0$	-2,0	0,0
Q_1	0, x	8,8	-2,0	0,0
R_1	0,0	0,0	0,4	0,0
S_1	0,-2	0,-2	-2,-2	4,0

- (a) Determine all pure-strategy Nash equilibria of game G .
- (b) Suppose game G is twice repeated (the discount factor is equal to 1) and players can observe the outcome of the first period before they make their choices in the second period.
- (b1) How many subgames does the twice repeated game have? Explain how you derive this number.
- (b2) Write down and explain the formula for determining how many pure strategies each player has in the twice repeated game.
- (b3) For which values of x is the following strategy (played by each player $i \in \{1, 2\}$) a subgame perfect Nash equilibrium in the twice repeated game?

Play Q_i in the first period. If (Q_1, Q_2) was played in the first period, play P_i in the second period. If (a, Q_2) , $a \neq Q_1$ was played in the first period, play R_i in the second period. If (Q_1, b) , $b \neq Q_2$ was played in the first period, play S_i in the second period. If (a, b) , $a \neq Q_1$, $b \neq Q_2$ was played in the first period, play P_i in the second period.

a.

BR yields

	P_2	Q_2	R_2	S_2
P_1	<u>4, 4</u>	<u>x, 0</u>	-2, 0	0, 0
Q_1	0, x	8, 8	-2, 0	0, 0
R_1	0, 0	0, 0	<u>0, 4</u>	0, 0
S_1	0, -2	0, -2	-2, -2	<u>4, 0</u>

Where overline is player 1's BR dependent on player 2's choice and vice versa

Implying that the pure-strategy Nash equilibria are

$$NE = \{(P_1, P_2); (R_1, R_2); (S_1, S_2)\}$$

b1)

In this game, we have imperfect information

Remember that "Every history defines a proper subgame"

This implies that there are 16 proper subgames (one from each node, for example one from P_1P_2 , one from P_1Q_1 so on) and one subgame as the whole game, so 17 subgames in the twice repeated game

b3

Strategy:

Both players play Q_i in the first period

If no player deviate, both play P_i in the second round

If player 1 deviates by playing anything other than Q_1 , then both play R_i in the second round

If player 2 deviates by playing anything other than Q_2 , then both play S_i in the second round

If both players deviate, then both play P_i in the second round

We then calculate the two-period normal form

For example, if (Q_1, Q_2) was played in the first round, no one deviated which implies that P_i is played in the second round. This gives payoff: $(8,8) + (4,4) = (12,12)$

Another example is: (Q_1, P_2) : Player 2 deviated. Therefore, S_i is played in the second round so $(0, x) + (4,0) = (4, x)$

And so on.

This implies the following normal form

$1 \backslash 2$	P_2	Q_2	R_2	S_2
P_1	8,8	4,4	2,4	4,4
Q_1	4,x	12,12	2,0	4,0
R_1	4,4	0,4	4,8	4,4
S_1	4,2	0,2	2,2	8,4

By underlining BR, it is clear that for $x \leq 12$, (Q_1, Q_2) is a pure strategy Nash equilibrium

However, if $x > 12$, the strategy (Q_1, Q_2) is not an equilibrium since players would deviate by playing P_i to get a higher payoff. So, if $x > 12$, no SPNE exists

Problem 2

The Prisoner's Dilemma below is played T times (with T a finite number).

1\2	C	D
C	3, 3	0, 4
D	4, 0	2, 2

Assume players' common discount factor is $\delta = 1$ and a player incurs the cost $\epsilon > 0$ in any period in which her action differs from her action in the previous period, i.e., when she switches her action from C to D or from D to C .

Show that the strategy pair in which each player chooses C at the start of the game and after any history that ends with (C, C) , and otherwise chooses D , is a subgame perfect equilibrium of the finitely repeated game when $1 < \epsilon < 2$. Use the one-stage deviation principle. (**Hint:** Any 'histories' in this game end in one of four possible outcomes of the stage game. Consider these in turn and determine if a player can profitably deviate from the strategy if the other player sticks to it.)

I consider deviations from Player 1 given that Player 2 adheres to her strategy

(C, C)

If Player 1 cooperates in any period, her payoff is a constant of 3.

If Player 1 cooperates in any period but the first, her payoff is $4 - \epsilon$ in the first period and 2 in every other period. Since $\epsilon > 1$ (c.f. the definition in the assignment) the deviation is not profitable

(D, C) or (D, D)

If Player 1 adheres to strategy D, her payoff is 2 in every period. If she deviates in the first period, we are in (C, D), her payoff will be $-\epsilon$ in the first period and $2 - \epsilon$ in every other period. Therefore, it is unattractive to deviate for Player 1.

(C, D)

Given that Player 1 adheres to her strategy, her payoff will be $2 - \epsilon$ in the first period and 2 in every other period.

If she deviates in the first period, her payoff is 0 in the first period, $2 - \epsilon$ in the second and 2 in the rest of the periods. Since $\epsilon < 2$, it is not attractive to deviate.

Problem 3

Anna (A) works as a financial consultant and can exert either high or low effort in managing the finances of her client Bertram (B). Bertram can choose whether to pay her a high fee or a low fee. Suppose that the following stage game is infinitely repeated:

$A \backslash B$	high	low
high	2, 3	0, 2
low	3, 0	1, 1

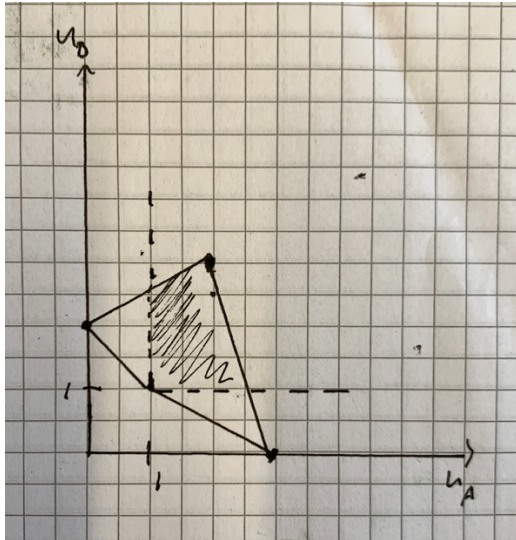
2

Anna and Bo have the same discount factor $\delta \in (0, 1)$.

- Draw the convex hull of average payoffs and indicate the average payoffs that can be supported in a subgame-perfect Nash equilibrium if δ is sufficiently close to 1.
- For what values of the common discount factor δ is (grim trigger; grim trigger) a subgame perfect equilibrium involving high effort and high fees in every period?
- Find an outcome path such that Anna's average payoff is close to 2.5.
- State a *strategy profile* that gives rise to the outcome path you suggested in part (c). For which values of δ is the strategy profile you suggested a subgame-perfect Nash equilibrium?

a.

Every payoff combination inside the convex hull is possible, and every average payoff in the marked area can be supported in a SPNE if δ is sufficiently close to 1, i.e., every point which yield a payoff greater than the NE outcome's (1, 1) for one party, or both



b.

Bertram earns his highest payoff in (H, H) , so he would never deviate

Anna would not deviate if

$$2 + 2\delta + 2\delta^2 + \dots = \sum_{t=0}^{\infty} 2\delta^t = \frac{2}{1-\delta} \geq 3 + \sum_{t=1}^{\infty} 1 \cdot \delta^t = 3 + \frac{\delta}{1-\delta}$$

So,

$$\frac{2}{1-\delta} \geq 3 + \frac{\delta}{1-\delta} \Leftrightarrow \delta \geq \frac{1}{2}$$

c.

$(H, H), (L, H), (H, H), (L, H), \dots$ has an average payoff of 2.5 to Anna

$$\begin{aligned} (1-\delta)(2 + 3\delta + 2\delta^2 + 3\delta^3) &= (1-\delta) \cdot \left(\sum_{t=0}^{\infty} 2\delta^{2t} + \sum_{t=0}^{\infty} 3\delta^{2t+1} \right) \\ &= (1-\delta) \left(\frac{2}{1-\delta^2} + \frac{3\delta}{1-\delta^2} \right) \\ &= \frac{2+3\delta}{1+\delta} \end{aligned}$$

Which is close to 2.5 if δ is close to 1

d.

(H, H) : Her kan Anna deviate til low således at hun får payoff 3 og derefter 1, 1, 1, ... Ved at holde sig til strategien får hun 2, 3, 2, 3, ... Det totale payoff bliver derfor:

$$\sum_{k=0}^{\infty} 2\delta^{2k} + \sum_{k=0}^{\infty} 3\delta^{2k+1} = \frac{2}{1-\delta^2} + \frac{3\delta}{1-\delta^2} \geq 3 + \frac{\delta}{1-\delta} \Leftrightarrow 0.37 \leq \delta < 1 \quad (1)$$

For Bo er der ikke incitament til at deviate i (H, H) , fordi han ved at deviate får 2 (fordi udfaldet så bliver (H, L) i stedet for 3).

Nu til (L, H) : Her giver det ikke mening for Anna at deviate, fordi hun ved at holde sig til strategien får 3 > 2.

Bo kan ved at deviate få 1, og så 1, 1, 1, ... herefter. Hvis han holder sig til strategien, får han fremtidige payoffs

$$0 + 3 \cdot \delta + 0 \cdot \delta^2 + 3 \cdot \delta^3 \dots$$

Derfor skal følgende ulighed være overholdt for at Bo ikke vil deviate:

$$\frac{0}{1-\delta^2} + \frac{3\delta}{1-\delta^2} \geq \frac{1}{1-\delta} \Leftrightarrow \delta \geq \frac{1}{2}$$

Det betyder, at for $\frac{1}{2} \leq \delta < 1$ er strategy profile $(H, H), (L, H), (H, H), (L, H)$ et SPNE